

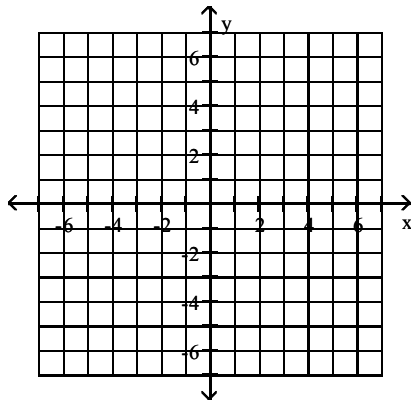
Name _____

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Plot the given point in a rectangular coordinate system.

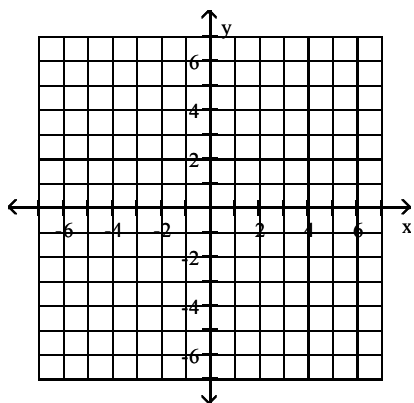
1) $(2, -3)$

1) _____



2) $(-6, -3)$

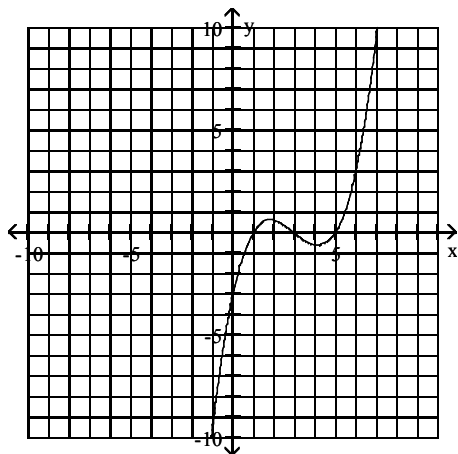
2) _____



Use the graph to determine the x- and y-intercepts.

3)

3) _____



Solve and check the linear equation.

4) $5x - 5 = 30$

4) _____

5) $-6x - 10 = -6 + 7x$

5) _____

6) $-7x + 5 + 5(x + 1) = -3x + 3$

6) _____

Solve the equation.

7) $\frac{x}{6} = \frac{x}{8} + 5$

7) _____

8) $\frac{x + 18}{6} + \frac{x + 9}{9} = x - 3$

8) _____

First, write the value(s) that make the denominator(s) zero. Then solve the equation.

9) $\frac{7}{x} = \frac{1}{2x} + 26$

9) _____

Use the five-step strategy for solving word problems to find the number or numbers described in the following exercise.

10) When 3 times a number is subtracted from 7 times the number, the result is 44. What is the number? 10) _____

11) When 10% of a number is added to the number, the result is 143. What is the number? 11) _____

Solve the problem.

12) A car rental agency charges \$150 per week plus \$0.20 per mile to rent a car. How many miles can you travel in one week for \$250? 12) _____

Solve the formula for the specified variable.

13) $d = rt$ for r 13) _____

Solve the equation by factoring.

14) $x^2 = x + 42$

14) _____

15) $3x^2 - 20x = 7$

15) _____

Solve the equation by the square root property.

16) $6x^2 = 54$

16) _____

17) $3(x - 9)^2 = 21$

17) _____

Solve the equation using the quadratic formula.

18) $x^2 + 5x + 1 = 0$

18) _____

19) $4x^2 = -8x - 1$

19) _____

Solve the polynomial equation by factoring and then using the zero product principle.

20) $x^3 - 10x^2 + 24x = 0$

20) _____

Solve the radical equation, and check all proposed solutions.

21) $\sqrt{x + 1} = 6$

21) _____

22) $\sqrt{3x + 28} = x$

22) _____

Solve the absolute value equation or indicate that the equation has no solution.

23) $|x - 8| = 4$

23) _____

24) $|4x - 2| - 5 = -12$

24) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Determine whether the equation is an identity, a conditional equation, or an inconsistent equation.

25) $2(2x - 10) = 4x - 20$

A) Identity

B) Conditional equation

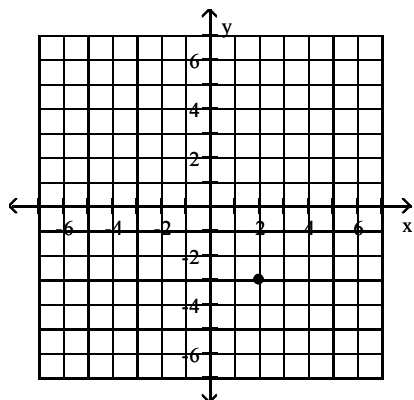
C) Inconsistent equation

25) _____

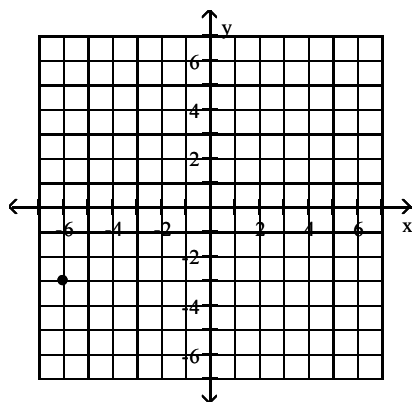
Answer Key

Testname: MTH_131_EXAM2_FALL_2012A

1)



2)



3) x-intercepts: 3, 1, 5; y-intercept: -3

4) $\{7\}$

5) $\left\{-\frac{4}{13}\right\}$

6) $\{-7\}$

7) $\{120\}$

8) $\left\{\frac{126}{13}\right\}$

9) $x \neq 0; \left\{\frac{1}{4}\right\}$

10) 11

11) 130

12) 500 miles

13) $r = \frac{d}{t}$

14) $\{-6, 7\}$

15) $\left\{-\frac{1}{3}, 7\right\}$

16) $\{-3, 3\}$

17) $\{9 \pm \sqrt{7}\}$

18) $\left\{\frac{-5 - \sqrt{21}}{2}, \frac{-5 + \sqrt{21}}{2}\right\}$

Answer Key

Testname: MTH_131_EXAM2_FALL_2012A

19) $\left\{ \frac{-2 - \sqrt{3}}{2}, \frac{-2 + \sqrt{3}}{2} \right\}$

20) $\{0, 6, 4\}$

21) $\{35\}$

22) $\{7\}$

23) $\{4, 12\}$

24) $\emptyset$

25) A